

ASSIGNMENT 4 - UNIT IV

MINI PROJECT

MOBILE PRICE ANALYSIS

Correlation Analysis, Visualization, Regression and Hypothesis Testing

Project Details	Information
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Abstract

This mini-project investigates how smartphone hardware specifications are associated with smartphone prices. A cleaned public smartphone dataset documented by Sarang Zambare contains device specifications such as LTE support, year, weight, display characteristics, processor speed, RAM, internal memory, camera, battery capacity and price. The repository states that the data were collected from GSMArena and that more than 1,000 smart devices had been compiled. For this project, an analysis sample of 82 phone records was extracted from the ordered dataset preview after excluding obvious tablets and large-screen non-phone devices. The analysis focuses on price in USD.

Correlation analysis shows that internal memory has the strongest positive relationship with price in the analytical sample ($r = 0.934$), followed by CPU speed ($r = 0.704$), while battery capacity is weakly related to price ($r = 0.128$). A simple linear regression using internal memory explains about 87.2% of the variance in price. A Welch two-sample t-test also indicates a statistically significant difference in mean price between phones with at least 4 GB RAM and those with less than 4 GB RAM ($p = 0.010$). These results suggest that storage, performance and premium hardware configurations are important price differentiators, while individual specifications should not be interpreted as causal determinants without controlling for brand and other confounding factors.

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1. Problem Statement

Smartphones are sold at widely different prices even when many of their specifications overlap. Consumers and manufacturers therefore need to understand which measurable specifications are most strongly associated with price. The objective of this project is to use statistical analysis to identify important price-related variables, visualize their relationships, build a regression model for price estimation, and test whether selected specification groups differ significantly in price.

Objectives

- Identify the variables most strongly correlated with smartphone price.
- Visualize distributions and relationships among price and specifications.
- Develop a linear regression model to estimate price from internal memory.
- Test whether phones with higher RAM have a different average price from phones with lower RAM.
- Interpret results carefully, distinguishing association from causation.

Research Questions

- Does internal memory have a strong relationship with smartphone price?
- Does RAM level significantly separate average phone prices?
- Which other numeric specifications show meaningful relationships with price?

2. Dataset Description

Source: Sarang Zambare, smartdevices_pricing, GitHub repository, cleaned CSV (data_clean.csv). The repository documents 15 fields and states that the source data were scraped from GSMArena. Prices were converted to USD using exchange rates stated in the repository as of 30 December 2018.

Variable	Meaning	Type
LTE	LTE support (1 = yes, 0 = no)	Binary
year	Year announced	Numeric
weight	Device weight in grams	Numeric
display_type	LCD, OLED or OTHER	Categorical
screen_size	Display size in inches	Numeric
ppi	Pixels per inch	Numeric
cpu_ghz	Processor speed	Numeric
ram	RAM in GB	Numeric
internal	Internal memory in GB	Numeric
camera	Rear camera resolution in MP	Numeric
front_cam	Front camera resolution in MP	Numeric
battery	Battery capacity in mAh	Numeric
price	Price in USD - target variable	Numeric

Analytical Sample

The extracted analytical sample contains 82 phone records. Obvious tablets or tablet-like entries (for example iPad, Iconia, Fire tablet, 1T and similar large-screen products) were removed to keep the analysis focused on phones. This filtering is intentionally transparent; the results describe this analytical sample and should not be generalized to every device in the full source dataset.

Statistic	Price (USD)	Internal Memory (GB)	RAM (GB)
Mean	298.17	62.24	2.42
Median	228.00	32.00	2.00
Std. Dev.	265.94	107.80	1.70

Statistic	Price (USD)	Internal Memory (GB)	RAM (GB)
Minimum	68.40	4.00	0.00
Maximum	1425.00	512.00	8.00

3. Data Preparation and Methodology

The analysis followed a standard exploratory and inferential workflow: inspect the source fields, restrict the sample to phone records, summarize numerical variables, measure pairwise Pearson correlation, visualize patterns, fit ordinary least squares regression, and apply Welch two-sample t-tests for group comparisons.

Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy and Statsmodels were used for data preparation, statistical testing, visualization and regression.

Key Statistical Measures

Pearson correlation coefficient: $r = \text{Cov}(X,Y) / (s_X s_Y)$. Values close to +1 indicate a strong positive linear association, values close to -1 a strong negative linear association, and values close to 0 little linear association.

Simple linear regression: $\text{Price} = \text{beta0} + \text{beta1}(\text{Internal Memory}) + \text{error}$. The slope beta1 estimates the average change in price associated with a 1 GB increase in internal memory in this sample.

Welch t-test: used to compare mean prices of two groups without assuming equal population variances. Statistical significance was evaluated at $\alpha = 0.05$.

4. Correlation Analysis

The correlation matrix indicates a very strong positive relationship between internal memory and price. Processor speed is also strongly associated with price, while RAM, PPI, weight and screen size show moderate positive associations. Battery capacity is comparatively weakly associated with price.

Variable	Correlation with Price (r)	Interpretation
internal	0.934	Strong positive
cpu_ghz	0.704	Strong positive
blth_v	0.634	Moderate positive
ppi	0.477	Moderate positive
ram	0.475	Moderate positive
weight	0.420	Moderate positive
screen_size	0.391	Weak positive
year	0.251	Weak positive
front_cam	0.236	Weak positive
LTE	0.223	Weak positive
camera	0.195	Weak positive
battery	0.128	Weak positive

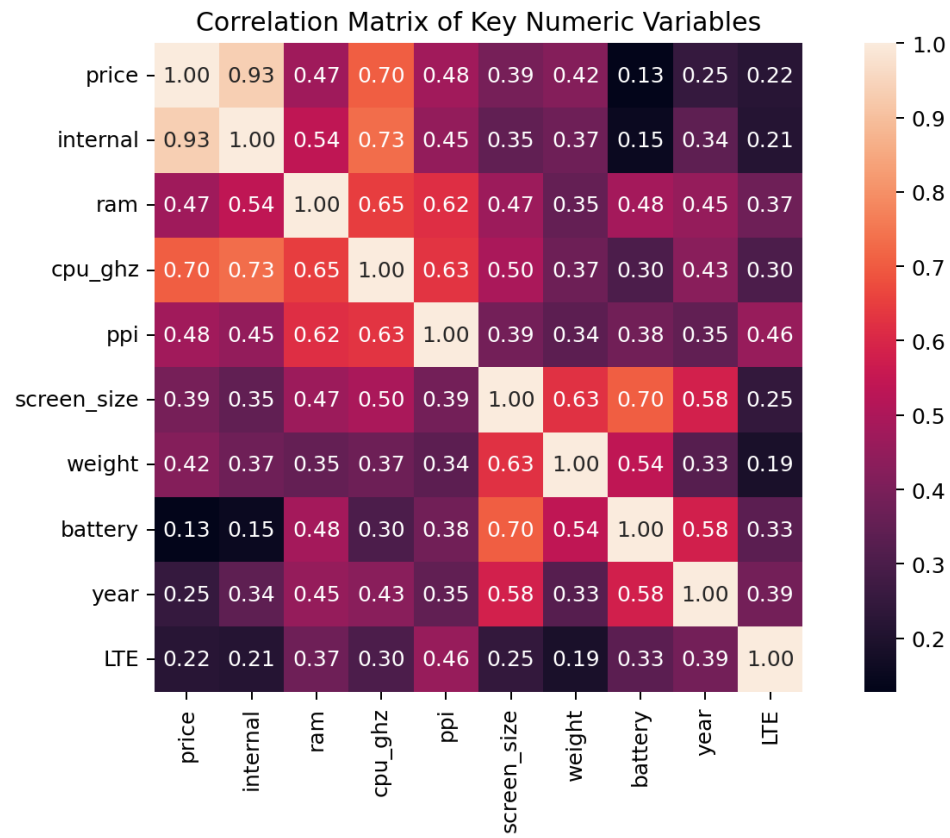


Figure 1. Correlation matrix of selected variables

Interpretation: internal memory is the dominant single numeric correlate in this sample ($r = 0.934$). This does not prove that increasing storage alone causes a price increase; higher-storage configurations are often bundled with higher-end processors, displays, cameras and brands. The correlation matrix is therefore best viewed as a screening tool for candidate predictors.

5. Visualization

The visual analysis supports the numerical results and makes the distributional characteristics easier to observe.

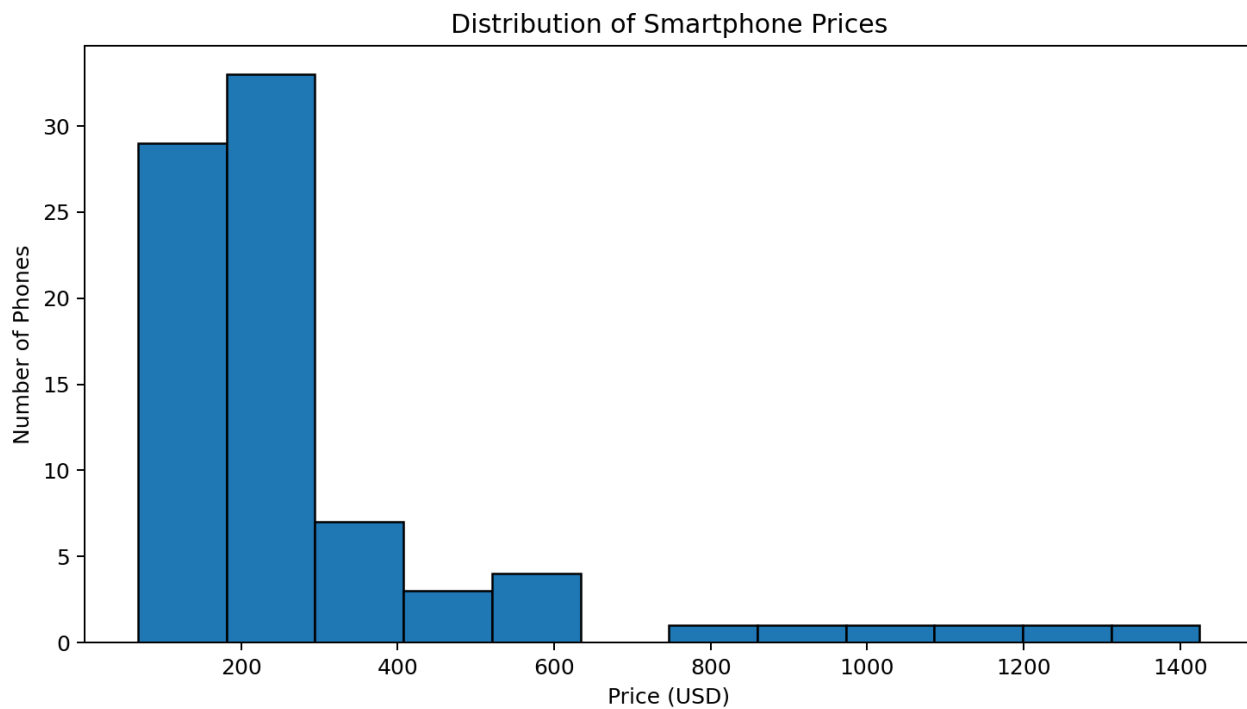


Figure 2. Distribution of smartphone prices

The price distribution is right-skewed because a small number of premium devices have prices far above the main cluster of phones. The mean price is therefore higher than the median.

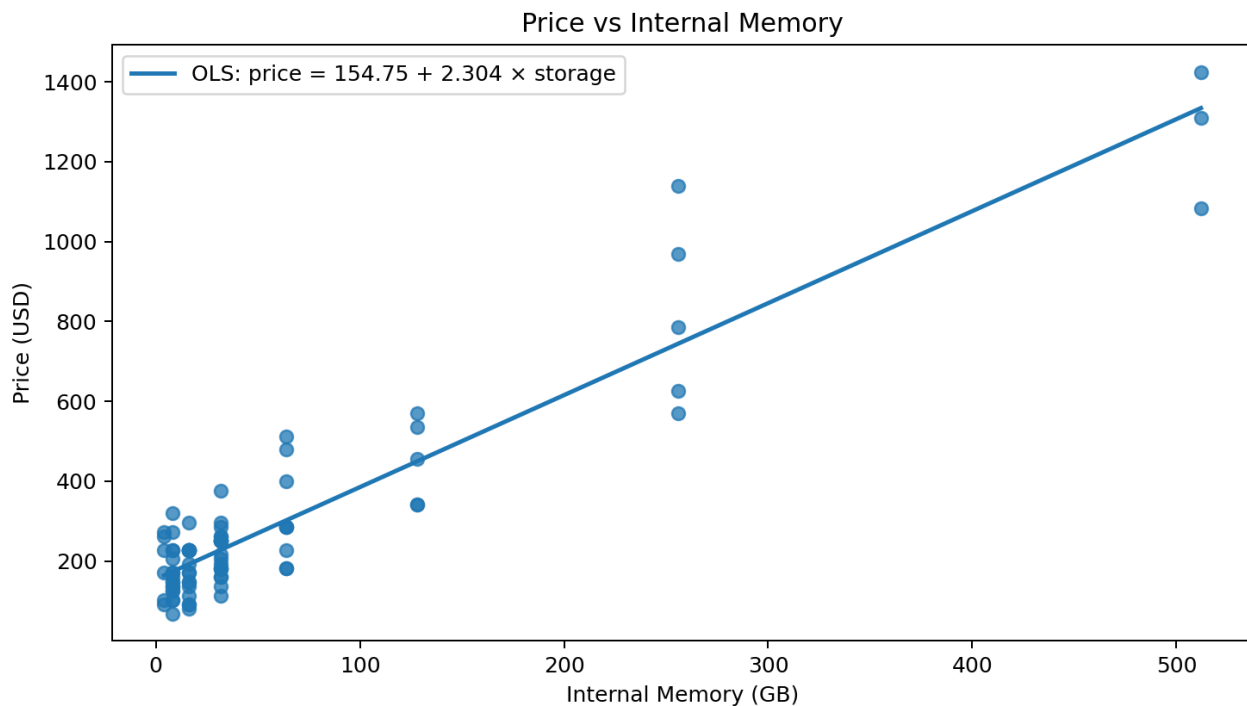


Figure 3. Price versus internal memory with fitted regression line

The scatter plot shows a strong upward trend. High-storage devices tend to occupy the higher-price region, although the points are not perfectly aligned, confirming that other specifications and brand effects also matter.

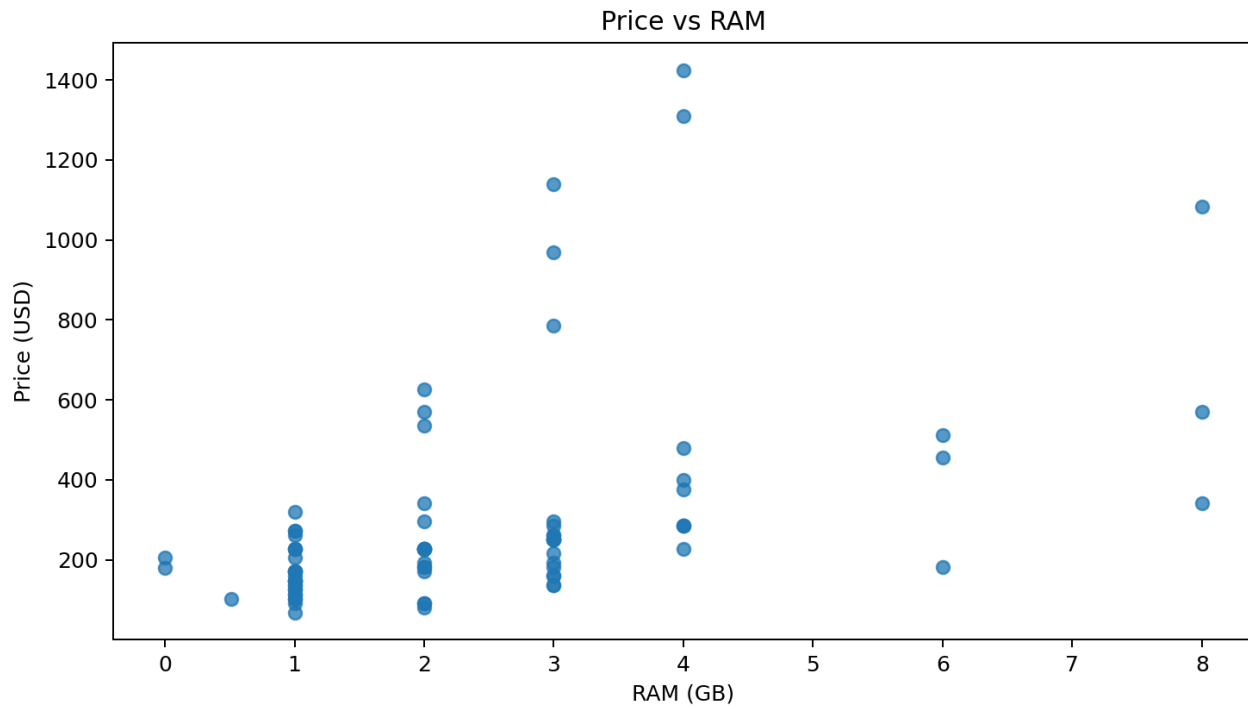


Figure 4. Price versus RAM

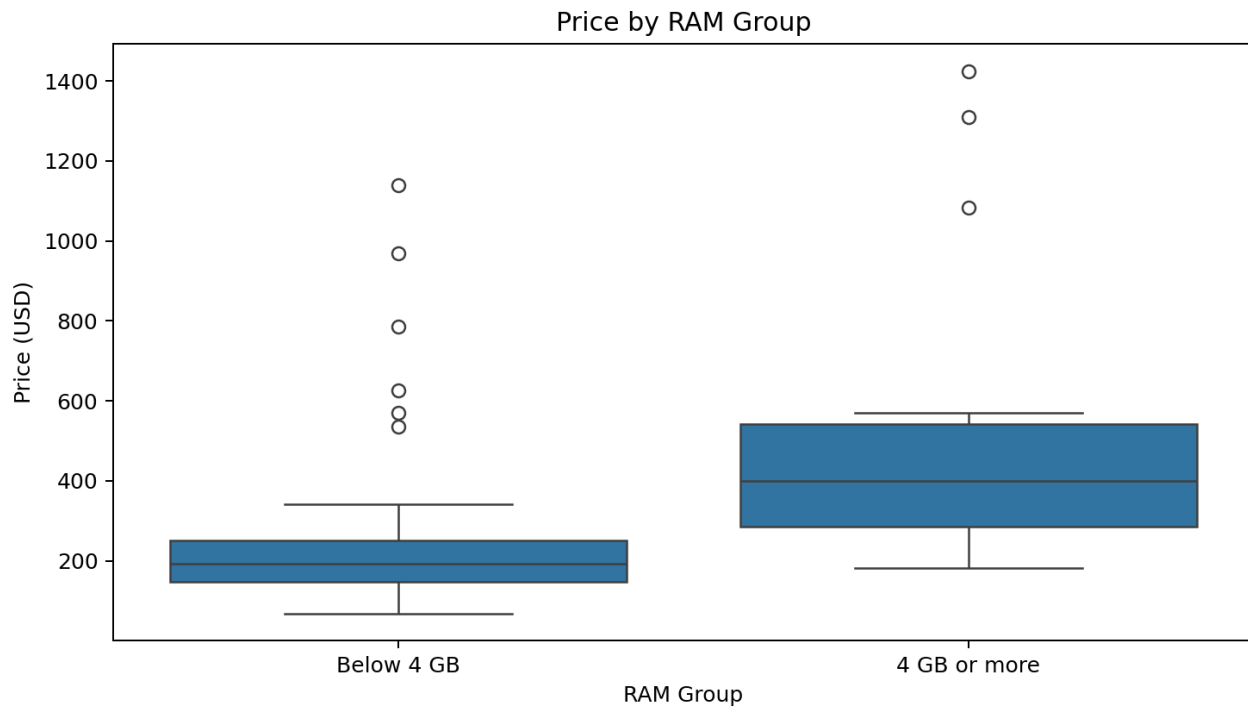


Figure 5. Price comparison by RAM group

6. Regression Analysis

6.1 Simple Linear Regression

The simple OLS model is: $\text{Price} = 154.75 + 2.304(\text{Internal Memory in GB})$. The model has $R^2 = 0.872$ and adjusted $R^2 = 0.871$. The slope is highly significant ($p < 0.001$).

Term	Coefficient	Std. Error	t	p-value
Intercept	154.754	12.216	12.67	8.22e-21
Internal memory (GB)	2.304	0.099	23.37	1.714e-37

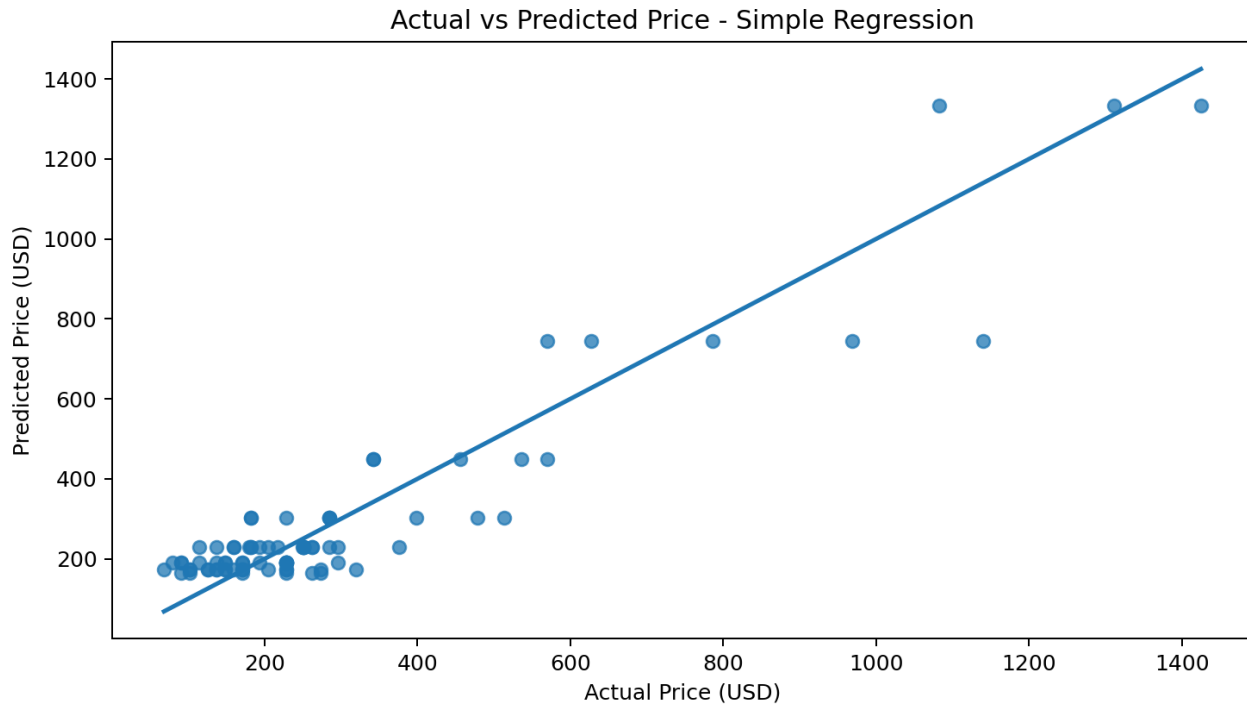


Figure 6. Actual versus predicted price for the simple regression

Interpretation: within this sample, a 1 GB increase in internal memory is associated with an average price increase of about \$2.30. The model explains 87.2% of observed price variation, which is high for a one-variable model. However, the result should not be treated as a causal effect because storage is correlated with other premium specifications.

6.2 Multiple Regression Robustness Check

A multiple linear regression was also estimated using RAM, internal memory, CPU speed, PPI, battery capacity, screen size, announcement year (centered at 2015) and LTE status. The model achieved $R^2 = 0.907$ and adjusted $R^2 = 0.897$, with an overall F-test p-value < 0.001 . Internal memory remained strongly significant ($p < 0.001$), while screen size and centered year were also statistically significant in this sample.

Predictor	Coefficient	p-value
RAM	-13.97	0.1061
Internal memory	2.27	4.404e-27
CPU speed	-4.41	0.9075
PPI	0.343	0.0500
Battery	-0.031	0.1319

Predictor	Coefficient	p-value
Screen size	101.18	0.0004
Year (centered)	-26.34	0.0043
LTE	33.61	0.2753

Caution: the multiple model shows signs of multicollinearity among some specification variables. Therefore, coefficient-level interpretations should be treated more cautiously than the overall predictive fit and the simple storage relationship.

7. Hypothesis Testing

7.1 RAM Group Comparison

Question: Do phones with at least 4 GB RAM have a different mean price from phones with less than 4 GB RAM?

H0: The two population mean prices are equal.

H1: The two population mean prices are different.

Group	n	Mean Price (USD)
RAM $\geq$ 4 GB	15	547.96
RAM < 4 GB	67	242.25

Welch t-test result: $t = 2.919$, $df = 15.48$, $p = 0.0103$. Since $p < 0.05$, H0 is rejected. The sample provides statistically significant evidence that mean prices differ between the two RAM groups.

7.2 Additional Display-Type Test

As a secondary check, an OLED-versus-LCD comparison was performed.

OLED mean = \$739.58 ($n=8$), LCD mean = \$258.41 ($n=70$). Welch $t = 2.486$, $df = 7.14$, $p = 0.0411$. This also crosses the 0.05 threshold, but the small OLED group makes the estimate less stable than the RAM comparison.

8. Results and Interpretation

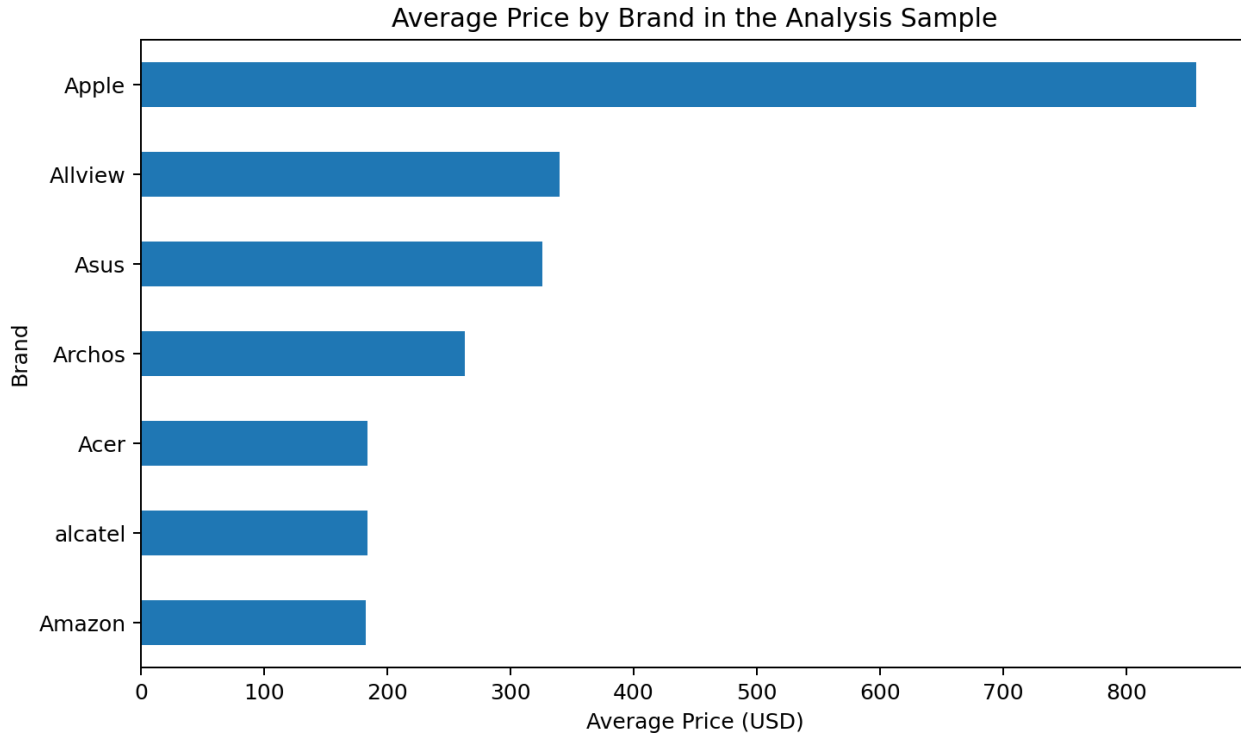


Figure 7. Average phone price by brand in the analytical sample

Finding	Evidence	Interpretation
Internal memory is the strongest single correlate	$r = 0.934$; simple regression $R^2 = 0.872$	Storage capacity is strongly associated with higher-priced device configurations.
Higher RAM group has higher mean price	4+ GB mean \$547.96 vs <4 GB mean \$242.25; $p = 0.0103$	The price difference is statistically significant in this sample.
CPU speed is strongly associated with price	$r = 0.704$	Faster processors are concentrated in more expensive devices.
Battery is a weak price correlate	$r = 0.128$	Large batteries are not necessarily a premium-price feature by themselves.
Premium brands dominate the upper end	Apple has the highest sample mean	Brand and product positioning likely contribute beyond raw hardware specifications.

A key interpretation point is that statistical association does not equal causation. The data are observational and cross-sectional, and several specifications move together. For example, premium phones tend to combine more storage, faster processors, better displays and higher brand positioning. A model that omits these interactions can overstate the apparent influence of any single feature.

9. Conclusion

The Mobile Price Analysis demonstrates how descriptive statistics, correlation, visualization, regression and hypothesis testing can be combined to understand a real-world pricing problem. In the analyzed sample of 82 phones, internal memory was the most strongly correlated numeric feature with price. A simple one-variable regression explained 87.2% of the observed price variation, and the RAM-group hypothesis test found a significant mean-price difference at the 5% level. These results suggest that premium hardware configurations are reflected in smartphone prices, while battery capacity alone is a weak indicator.

For future work, the analysis could be extended to the complete cleaned dataset, include manufacturer/brand as categorical predictors, use regularization or tree-based models, test interaction terms, and perform train-test validation. Such extensions would improve predictive reliability and reduce the risk of interpreting correlated specifications as independent causal drivers.

10. References

- Zambare, S. smartdevices_pricing. GitHub repository. Data collected from GSMArena; cleaned dataset: csv/data_clean.csv. https://github.com/sarangzambare/smartdevices_pricing
- GSMArena. Mobile device specifications source referenced by the dataset repository. <https://www.gsmarena.com/>
- Python scientific computing stack used for analysis: Pandas, NumPy, Matplotlib, Seaborn, SciPy and Statsmodels.

Appendix A - Sample Records

The table below shows an excerpt of the analytical sample. The complete machine-readable analytical subset is also provided as mobile_price_analysis_sample.csv alongside this report.

Model	Year	RAM	Storage	CPU GHz	Screen	Battery	Price
Acer Liquid Z6 Plus	2016	3.0	32	1.30	5.50	4080	285.0
Acer Liquid Z6	2016	1.0	8	1.25	5.00	2000	136.8
Acer Liquid X2	2015	3.0	32	1.30	5.50	4020	262.2
Acer Liquid Zest	2016	1.0	8	1.30	5.00	2000	125.4
Acer Liquid Jade Primo	2015	3.0	32	1.40	5.50	2870	250.8
Acer Liquid Z330	2015	1.0	8	1.10	4.50	2000	125.4
Acer Liquid Z320	2015	1.0	8	1.00	4.50	2000	102.6
Acer Liquid Z630	2015	1.0	8	1.30	5.50	4000	228.0
Acer Liquid Z530	2015	1.0	8	1.30	5.00	2420	171.0
Acer Liquid Jade Z	2015	1.0	8	1.50	5.00	2300	228.0
Acer Liquid Z520	2015	1.0	8	1.30	5.00	2000	148.2
Acer Liquid Z220	2015	1.0	8	1.20	4.00	1300	102.6